

From radiobiology to radiotherapy: dose homogeneity in cells after alpha irradiation in measurements and Monte Carlo simulations

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Abstract

Introduction: Proton radiotherapy offers an advantage in sparing healthy tissue compared to photon therapy due to the specific interaction of protons with the patient's body. In radiobiological experiments, alpha sources are commonly used instead of proton accelerators for convenience, but ensuring a uniform dose distribution is challenging. Properly designing the cell irradiation setup is crucial to reliably measure the average cellular response in such experiments. The objective of this research is to underscore the importance of dosimetric validation in radiobiological investigations. While Monte Carlo (MC) simulations offer valuable insights, their accuracy needs experimental confirmation. Once consistent results are obtained, the reliance on simulations becomes viable, as they are more efficient and less cumbersome compared to experimental procedures.

Material and methods: The simulations are performed with three MC code-based tools: Geant4-DNA, GATE, and SRIM to model the alpha radiation source and calculate dose distributions for various cell irradiation scenarios. Dosimetric verification of the experimental setup containing a ²⁴¹Am source is performed using radiochromic films. Additionally, a clonogenic cell survival assay is carried out for the DU145 cell line.

Results: The study introduces a novel source simulation model derived from dosimetric measurements. The comparison between dosimetric results obtained with simulations and measured experimentally yields a gamma (3%/3mm) parameter value exceeding 99%. Furthermore, the LQ model parameters fitted to survival data of DU145 cells irradiated with particles emitted from ²⁴¹Am source demonstrate consistency with previously published findings.

Conclusions: Radiobiological experiments investigate cellular responses to various irradiation scenarios. Challenges arise with densely ionizing radiation used in clinical practice, particularly in ensuring uniform dose delivery for reliable experiments. MC codes aid in simulating dose distribution and designing irradiation systems for consistent cell treatment. However, experimental validation is essential before relying on simulation results. Once confirmed, these results offer a cost-effective and time-efficient approach to planning radiobiological experiments compared to traditional laboratory work.

Keywords: Monte Carlo simulations; cell survival; dosimetry; radiation biology; high LET radiation.

Introduction

Most studies on the impact of ionizing radiation on cell populations have used X-ray or gamma radiation as the interacting factor. X-rays can be easily handled with a variety of irradiation devices and deliver, directly measurable, uniform doses to the target. Unlike low linear energy transfer (LET) radiation, α -particles are difficult to use because they are slowed down before reaching their target. This dose is difficult to measure and must be calculated. The study of the interaction of

α -particles with biological material is justified by the fact that modern radiotherapy increasingly uses heavy charged particles such as protons and carbon ions. Proton radiotherapy is considered the most advanced form of radiotherapy. It should be taken into account that the number of patients who have been treated with proton therapy reached 1.7 million worldwide in 2020.¹ Alpha particles interact similarly to protons and heavy ions, and the effects of their interaction differ significantly from those caused by X-ray and gamma radiation. Therefore, we have important reasons to use α -particles as an interacting agent

despite the difficulties associated with this type of irradiation.² When the average dose to a cell culture is low, α -particle and proton irradiation does not distribute the dose uniformly, but rather a small number of cells can receive a large amount of energy, while most cells receive no dose at all.³

Ionizing radiation causes many types of DNA damage, including double-strand breaks (DSB) and single-strand breaks (SSB). DNA damage occurs in cells as a result of exposure to ionizing radiation. Of these, DSB is the most mutagenic and lethal because it is difficult to repair, while improperly repaired DNA damage can trigger carcinogenesis.⁴ The cellular response to ionizing radiation is of fundamental importance in radiobiology because we are constantly exposed to environmental or medical radiation. Studies of α -particle irradiation of cells play an important role in understanding the biological effects caused by low doses of high-LET radiation and in studying the dependence between dose and biological response that forms the basis of radiotherapy.

The alpha source selected in our study is ^{241}Am . In the Radiobiological Laboratory of the Biomedical Physics Division at the Faculty of Physics of the University of Warsaw, we have built an alpha particle irradiator whose task is to study cellular responses to alpha emission. In general, cellular doses within the Petri dish may be heterogeneous (non-uniform) due to the flat shape of the disk source placed parallel to and separated from the cells by a thin layer of air and Mylar foil. The isotropic alpha particle emission from the planar disc source also plays an important role. Our commercial α -emitting source consists of radioactive foil about two micrometers thick. The non-uniform thickness of the film leads to non-uniform activity distribution on the surface of the disk source, therefore the fluence coefficient of alpha particles towards the target cells in the culture flask would also become non-uniform. Previous estimates of the dose rate distribution assumed that the activity concentration of our alpha source was uniform on its surface.⁵ Still, measurements performed using Gafchromic EBT3 film show the non-uniform thickness of the ^{241}Am layer. Dosimetry and its associated experimental uncertainties are essential in interpreting radiation effects, especially when high LET ionizing radiation is used and their relative biological effectiveness must be estimated. In research on particle radiobiology, certain difficulties result mainly from the small distances of particles in matter. However, due to the ability to create thin cell monolayers, precise information about the dose distribution is necessary.

Our goal is to provide increased dosimetric accuracy, which is of great importance in the correct interpretation and quantification of the effects of α -irradiation.⁶ Exposure of cell culture to alpha particles in vitro is performed in the Biomedical Physics Division. The alpha irradiation is operated using a ^{241}Am alpha-emitting disk source. The exposure of alpha particles to cells in a culture dish of different sizes was characterized by analytical calculations and Monte Carlo

simulations, assuming a heterogeneous (non-uniform) distribution of ^{241}Am activity in the disk source. The Gafchromic EBT3 film is a film-type dosimeter, which allows the estimation of the radiation dose applied to the film by measuring its optical density. EBT3 film has been widely used to measure the dose of photon, proton, and electron beams, but not to measure low energy α -particles. To measure dose uniformity, we use EBT3-unlaminated films for alpha dosimetry and further suggest that EBT3-unlaminated film can be used to check the uniformity of produced alpha emission sources. Here we describe a simple experimental device that was designed and built for irradiation of cell monolayers. The system uses particles from a ^{241}Am source emitted in the air. A Mylar exit window was used for irradiation. The distance from the source to the output can be modified according to the requirements of the experiment, resulting in different doses. There is no need to use honeycomb collimators. For α -dosimetry, Monte Carlo calculations are proposed for the stopping power $S(E)$ and range $R(E)$ of particles in liquid water as a function of their energy E , according to our previous publication.⁵ This article describes a simple method for irradiating cell cultures with alpha particles, provides related dosimetry, and provides an example of effects observed in cell populations.

Material and Methods

Tools

MC simulations

Three frameworks (Geant4-DNA, GATE, and SRIM) based on Monte Carlo methods were used in this work. To compare the accuracy of these sets of codes, the range of alpha particles of energy equal to 5 MeV was calculated. Simulations were performed for three setups containing water or water and air, which are shown in **Figure 1**.

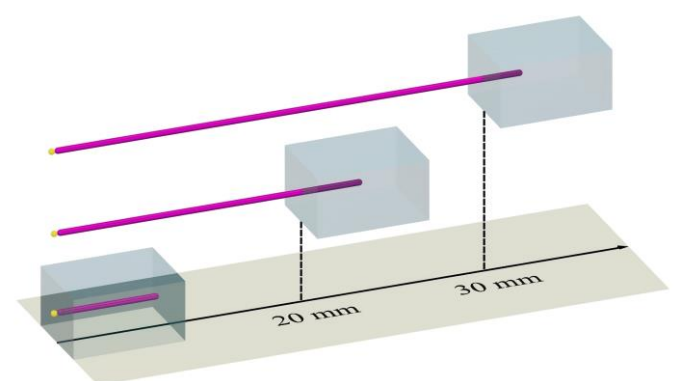


Figure 1. Irradiation setups used in simulations performed with Geant4-DNA, GATE, and SRIM to calculate the range of alpha particles. The water phantoms are 50 μm long cubes. The point source of alpha radiation is marked as a yellow sphere. The violet lines represent the direction of alpha particle movement.

Geant4 is a freely available set of tools for simulating and tracking particles in matter. Various physical models, found in different iterations of Geant4, are utilized for high LET particles.⁷ In Geant4 (ver. 10.07.p01) simulations, the QGSP_INCLXX and Geant4_DNA physics lists were used. The Geant4_DNA list was used in simulations regarding the range of alpha particles in water.⁸⁻¹⁰ This list includes the processes of nuclear scattering, electronic excitation ionization, and charge decrease and increase. The use of these cross-sections is limited to liquid water only.¹¹ The simulations of dose distributions were performed using the QGSP_INCLXX list, which is used for hadrons simulations.

GATE (Geant4 Application for Tomographic Emission) is a nuclear medicine simulation toolkit¹² that in general may simulate the transport of particles in matter. In this paper, version vGATE9.2 was used with a high-precision QGSP_BERT_HP physics list applied.

SRIM (Stopping and Range of Ions in Matter) is a software package used for studying in the field of ion beam physics and material science.¹³ The whole package contains a Monte Carlo calculation tool, called TRIM, which simulates interactions of ion beams with materials.

Due to limitations in SRIM (only linear monoenergetic beams can be applied) and the fact that GATE is based on Geant4 codes, these software were used only for calculations of the range of alpha particles in water.

Alpha source

The flat source of alpha radiation was ²⁴¹Am manufactured by Eckert & Ziegler, Prague, Czech Republic with a 50 mm diameter and an activity equal to 1.96 MBq. The active layer, where alpha particles were produced, consists of pure gold mixed with AmO₂ in a ratio of 1:0.0425 and thickness less than 1 µm. According to manufacturer information, the layer of gold covering the active layer has a thickness of (2.0 ± 0.2) µm. The simulated energy spectrum of ²⁴¹Am with a half-life of 432.6 years included alpha particles of energies 5.48 MeV (84.5%), 5.44 MeV (13.1%), 5.39 MeV (1.6%), 5.54 MeV (0.3%), and 5.51 MeV (0.2%), with a weighted energy average of 5.46 MeV.¹² Throughout the decay process, there is an emission of low-energy gamma radiation. Nearly 77% of all of the photons emitted during the decay of ²⁴¹Am occur in the range of 11.9 to 69.8 keV.¹⁴ This radiation results in minimal dose absorbed in cells (dose rate less than 50 mGy/min), with negligible biological effects. Consequently, it was determined that further simulations and considerations would exclude this type of radiation. The place of alpha particle emission in the active layer (as a result of the decay of ²⁴¹Am) was modeled based on the random position of ²⁴¹Am atoms in the active layer. It was done based on Gaussian distribution using Geant4.

Dose and energy measurements

Dosimetric verification of the ²⁴¹Am source was performed with Gafchromic® EBT3-Unlaminated film (EBT3; #11022001P1).

Gafchromic films were used to measure the dose and its spatial distribution for all irradiation setups described in the next sections. Films were scanned using Epson Expression 12000XL flat-bed scanner (Epson, Suwa, Japan) in transmission mode and at a resolution of 600 dpi. Triple channel red-green-blue (RGB) format was used with each pixel having a color depth of 48 bits (16 bits per channel). Images were saved in tagged image file format (tiff). For film calibration a flat source of ²⁴¹Am (AM1AP1 0053U, Eckert & Ziegler) with a dose rate of 554 mGy/h was used. Measurements were performed in a vacuum chamber with radiochromic films, which were placed under a 1 cm diameter steel collimator located 8 cm from the americium source and irradiated with doses from 0 to 6 Gy.

A silicon semiconductor detector boron implanted was used to measure the energy of particles from the ²⁴¹Am source. The detector, with a total area of 300 mm², was located in a chamber connected to a vacuum pump and a vacuum gauge (PWN100/PN100 by Unifra Unima, Poland), which allowed it to measure the pressure inside the chamber up to 5 Pa (or 0.05 mbar). The vacuum obtained is necessary due to the low range of alpha particles in the air. The detection system used for energy measurement and radiochromic film irradiation is shown in **Figure 2**.

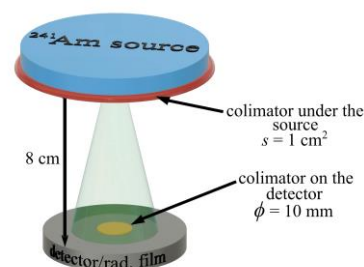


Figure 2. Setup geometry including ²⁴¹Am source in a vacuum chamber with the Si-detector and radiochromic film. Setup was used for either measurements or simulations.

Irradiation setups

Energy spectrum analysis

To estimate the energy spectrum of alpha particles emitted from a flat ²⁴¹Am source, simulations with Geant4 and measurements with Si-detector were performed. In the Geant4 simulation, the vacuum chamber with a Si-detector was built. The measurement setup (**Figure 2**) includes the alpha particle source and the detector, both with the collimators (the one under the source with a surface of 1 cm² and the second with a diameter of 1 cm on the detector). The diameters of the collimators were selected to obtain the appropriate number of particles reaching the detector, taking into account its dead time. The alpha particles were generated randomly over the entire surface of the source and to depth according to the Gaussian distribution with mean $\mu = 1.29$ µm and standard deviation $\sigma = 0.26$ µm, but not more than 3 σ . The initial energy of the alpha particles was randomised from

the three main energies of americium decay. The particles pass through the gold layer covering the radioactive isotope, where they reduce energy and smear the energy spectrum.

Dose homogeneity studies

To examine the dosimetry of the alpha particle source, a series of measurements and simulations were carried out in various irradiation configurations. Each setup differed in distance between the source and the radiochromic film, absorber applied, or motion of the source.

A radiochromic film was positioned above the fixed source in static irradiation setups (**Figure 3**). The films were irradiated without any absorber (placed directly on the source) and with an absorber: covered by 3 μm of Mylar foil and placed at a distance of 5 mm above the source. For each setup, an exposure time was equal to 133 seconds resulting in dose of (1.27 ± 0.14) Gy and (1.023 ± 0.081) Gy, respectively. For comparison, Monte Carlo simulations using the GATE and Geant4 tools were conducted. In the simulation, the world was filled with air, the active layer of the radiochromic film had a thickness of 14 μm , a density of 1.2 g/cm³, and was built with 56.8% of hydrogen, 27.6% of carbon, 13.3% of oxygen, 1.6% of aluminum, and 0.6% of lithium.¹⁵

The dynamic experimental setups were achieved using a laboratory shaker with the source located on the flask platform (marked as a dark red cube) and radiochromic films mounted above the source to a metal arm. The source was elliptically moving during the irradiation in a plane parallel to the film fixed above. In the first configuration, a radiochromic film was in constant contact with the surface of the source when shaking. The next two configurations included the use of an absorber: 5 mm of air and 5 mm of air together with 3 μm of Mylar. In these scenarios, the Mylar foil was attached to a metal arm, shown in **Figure 4**. Analogously to measurements using static setups, the exposure time was 133 seconds.

Statistical analysis

The gamma index (γ) is one of the most frequently used indicators to verify the compliance of the received dose in radiotherapy. The gamma index combines dose and distance differences to calculate a dimensionless metric for each point in the evaluated distribution. The format $dD(\%)/dr(\text{mm})$, where D is the dose and r is the distance, is the standard for reporting the criteria for passing the gamma index. The most common passing criteria used is 3%/3mm.¹⁶

To estimate the homogeneity of dose distribution, the coefficient of variation (CV) was calculated. CV expresses the relative variability of the population in relation to the mean. It is defined as the ratio of the standard deviation σ to the mean μ ($CV = \sigma/\mu$). In this case, μ was the average pixel value within the irradiated area and σ was its standard deviation.

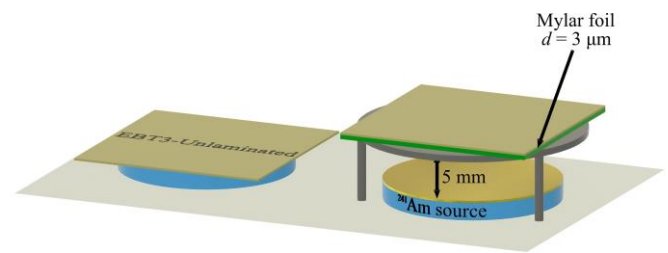


Figure 3. Static irradiation setups: radiochromic film is placed directly on the source (left) or moved 5 mm above the source and covered by Mylar foil (right).

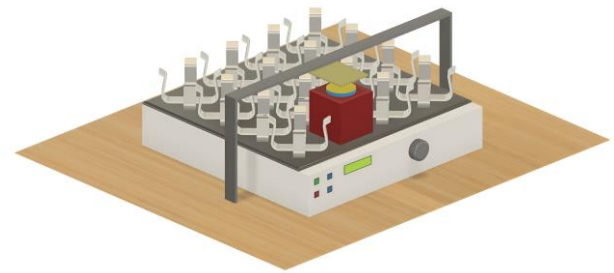


Figure 4. Dynamic experimental irradiation setups: a laboratory shaker with the source located on the flask platform (marked as a dark red cube) and radiochromic films mounted above the source to a metal arm.

To obtain the best curve fitting parameters, the Least Squares Method was used. After fitting the curve to the data and obtaining the parameters with uncertainties, the residuals were calculated. Once the residuals have been calculated, the determination of the corresponding χ^2 value of goodness of fit was obtained. If the model describes the data and the estimated uncertainty of each point has been determined, the reduced χ^2 value is approximately equal to 1. Statistical analysis was conducted utilizing a Python script employing the scipy.stats library.

Cell survival studies

Cell culture

The DU145 (HTB-81) cell line was purchased from the American Type Culture Collection (ATCC). DU145 cells were grown in a Dulbecco's modified Eagle medium (DMEM-F12, biowest, Cat #L0092-500) with a final concentration of 10% fetal bovine serum (FBS, biowest, Cat #S1810-500) and 0.6% penicillin-streptomycin (AppliChem GmbH, Cat #A8943). Cells were routinely cultured in a 75 cm² flask as a monolayer, with medium exchanges every 2 to 3 days, and passaged with 2 mL of 0.25% trypsin-EDTA solution (Biowest, Cat #X0930-100) when cells reached 80-90% confluence. Neutralisation of the trypsinization process was accomplished by adding 7 mL of complete growth medium to the trypsinized cell suspension. Cell cultures were routinely incubated at 37°C, in air with 5% CO₂.

Irradiation

Three days before irradiation, cells were seeded onto glass coverslips which were then placed in a 6-well plate. Cells were seeded at a density of 1.6×10^5 cells/well and then counted using automated cell counter EVE™ (NanoEnTek Inc.). Cells prepared in this way were irradiated with alpha radiation emitted by a flat source of ^{241}Am with a dose rate of 7.37 mGy/s, which corresponds to 1 Gy per 136 seconds. Each glass coverslip was 30 mm in diameter and was placed inside a Petri dish (93 mm in diameter). Then, the 4 μl of medium (corresponds to approximately 5.7 μm thickness over the coverslip) was added and in the next step, the cells were covered with 6 μm mylar foil (Figure 5). After covering the cells with mylar foil, the source of alpha particles was put directly on top. The Petri dish was closed and placed into the incubator for a given amount of time according to the estimated dose rate. The prostate cancer cells were irradiated to absorbed doses equal to 0, 0.16, 0.32, 0.65, 1.31, and 2 Gy.

Clonogenic survival assay

Directly after irradiation, DU145 cells were set up for the clonogenic survival assay. In the first step, cells were trypsinized and counted. Then, the cells were plated in Petri dishes containing 10 mL of medium. Cells were seeded at a density of 3,000 cells/dish in the control panel and of 5,000 to 50,000 cells/dish in the irradiated panel. Following seeding, cells were incubated for 10 days to allow colony formation. Colonies were stained with Giemsa solution. The number of colonies was investigated with countPHICS software.¹⁷ Comparing the number of colonies formed to the number of cells seeded allows us to determine the plating efficiency (PE). Surviving fractions of irradiated cells were calculated regarding the PE of control cells kept at the same conditions as irradiated cells. Four replicates for each dose and three for control cells were investigated within a single experiment. There were two experiments performed independently. To describe the dose-effect curve, the linear-quadratic (LQ) model was used, which relates the survival to dose as $\text{SF} = \exp(-\alpha D - \beta D^2)$, where α and β , parameters describing the cell radiosensitivity, are constant

and D is the absorbed dose. The degree of slope of the curve, described as the α/β ratio - this unit of Gy, illustrates the radiosensitivity of cells. Thus, cells with high α/β ratios see a relatively constant rate of cell killing with increasing dose, while those with a low α/β ratio show a pronounced curvature. It corresponds to the two contributions to cell killing. The α parameters express death from 'single hit' events - that is, lethal damage caused by a single incident particle (high LET radiation), while the β parameters reflect 'multiple hit' (low LET radiation). Since cell exposure was performed with high LET radiation, the mono-exponential curve was fitted. As a consequence, exposure to high LET radiation (alpha particles), where the direct interaction is dominant, the survival curves are described as mono-exponential dose-dependent.

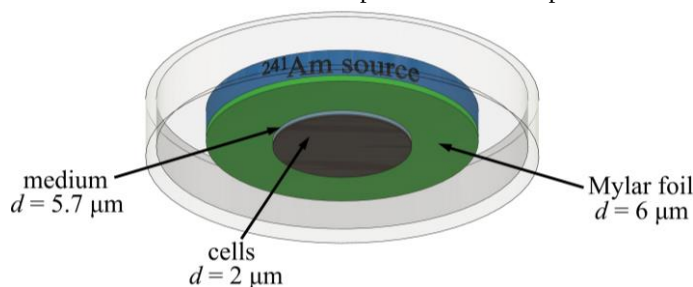


Figure 5. Experimental setup to irradiate DU145 cells with alpha radiation (view from below).

Results

Characteristics of alpha particles

Using three available simulation tools (Geant4-DNA, GATE, and SRIM), the range of the alpha particle with the initial energy of 5 MeV for various measurement setups was calculated. The results are presented in Table 1.

The values obtained prove the possibility of using these three simulation tools equally. For each setup, all values are in agreement within three standard deviations of the mean, thus based on the 3σ rule no significant difference can be found.

Table 1. The range of alpha particles in three measurement setups shown in Figure 1.

Simulation tool	Range in water (μm)		
	water only	20 mm of air + water	30 mm of air + water
Geant4-DNA	37.62 ± 0.40	16.54 ± 0.27	6.29 ± 0.27
GATE	36.97 ± 0.20	16.37 ± 0.38	6.40 ± 0.40
SRIM	36.49 ± 0.25	16.40 ± 0.40	6.20 ± 0.36

The energy spectrum of ^{241}Am

Model of the source according to the manufacturer

Based on the manufacturer's parameters, simulations of the spectrum of alpha particles reaching the Si-detector were made. Both energy spectra, simulated and measured, are shown in **Figure 6**.

For both spectra, the two-sample Kolmogorov-Smirnov test was performed. The obtained simulation results do not agree with the measurement results. While the peak position is almost at the correct location, the height of the simulated spectrum is lower than the measured one. Therefore a new source model should be proposed which will be described in the next section.

In subsequent simulations, the center of the active layer was placed at different depths from 0.8 to 2 μm below the source surface and then the average energy of the obtained peak was read. The results are presented in **Figure 7a**. For the average energy (equal to 4902 keV) measured with the Si-detector, the location of the center of the active layer was read at a depth of 1.29 μm . The energy distribution was still flattened, this is due to the uniform generation of alpha particles at a thickness of 0.4 μm . The solution to this problem was to change the generations of particles from uniform to generation in thickness with the probability resulting from the normal distribution with mean value $\mu = 1.29 \mu\text{m}$ and standard deviation $\sigma = 0.26 \mu\text{m}$, but not more than 3σ .

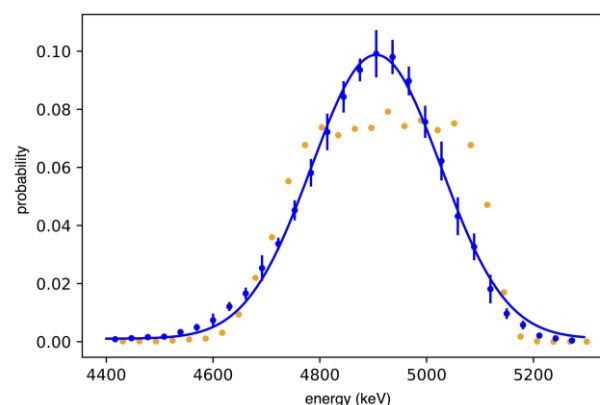


Figure 6. The energy spectrum of alpha particles emitted from the ^{241}Am measured with a Si-detector (blue points) and simulated using Geant4 based on source characteristics given by the manufacturer (orange points). Error bars represent the square root of the number of counts.

Proposed model of the source

A comparison of measured and simulated spectra is shown in **Figure 7b**. Assuming equality of variances of the presented distributions, a t-test for the difference between the mean values in two different groups was performed. For a confidence interval of 95%, there is no reason to reject the hypothesis of means equality.

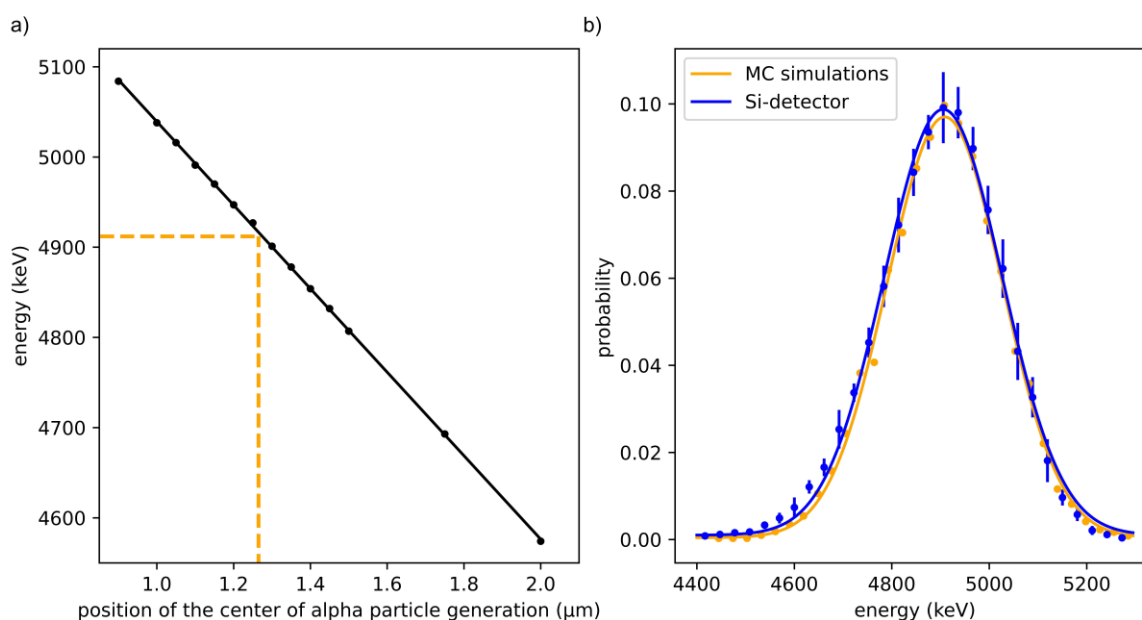


Figure 7. The mean value of energy as a function of the position of alpha generation in an active part of the source (a) and comparison of the measured and simulated with this method energy spectrum (b).

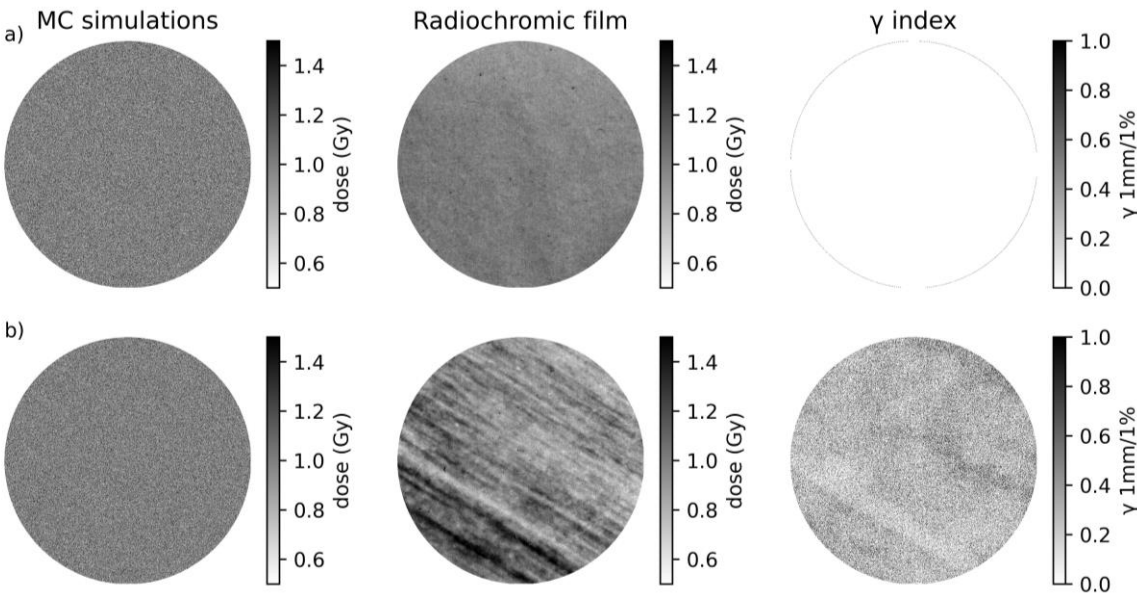


Figure 8. Comparison of dose distribution simulated with Geant4 (left) and measured with radiochromic films (middle). The gamma parameter (right) was calculated for irradiation setup with radiochromic film placed directly on the source (a) and covered by 3 μm of Mylar foil and moved 5 mm above the source (b).

Dose distributions

The spatial dose distributions and the dose values are presented in **Figure 8**. Based on dose distributions, measured and simulated, gamma parameters were calculated for two different setups described in **Figure 3**.

In the presented simulations and measurements, the absorbed dose was equal to 1 Gy. For gamma parameter settings of 1%/1mm, a compatibility of 77.67% was obtained in the case with setup with radiochromic film placed directly on the source and 99.84% in the second case. Minor inconsistencies in the second case result from different methods of writing data on the outskirts of the circle in the simulations and reading from the radiochromic films.

Homogeneity of dose distribution

Based on the irradiated scanned radiochromic films, CV values were calculated for five irradiation setups and listed in **Table 2**. Mean values and their standard deviations were calculated using two radiochromic films irradiated under the same conditions.

As described in the paragraph on statistical analysis, the coefficient of variance indicates the homogeneity of the absorbed dose. For the film placed directly on the source, its calculated value is the highest, suggesting the least homogeneous distribution of the dose. The two methods proposed resulted with the expected effect. Both moving the film away from the source and moving the source during irradiation improved the quality of the results. However, it should be noted that the values listed in **Table 2** were calculated for the same irradiation time, but different doses. For this reason, the CV is higher for setups with absorbers than for setups without.

Table 2. Calculated values of CV for dynamic irradiation setups compared with static ones.

Radiochromic film placed	CV
directly on the source	0.1163 ± 0.0073
5 mm above the source (covered by Mylar foil)	0.0808 ± 0.0055
directly on the source in the shaker	0.0693 ± 0.0071
5 mm above the source in the shaker	0.0730 ± 0.0011
5 mm above the source (covered by Mylar foil) in the shaker	0.0937 ± 0.0013

Table 3. Survival fraction data for DU145 cells irradiated with 2 Gy of alpha particles. The alpha parameter of the LQ model with corresponding survival fraction for 2 Gy calculated based on the fit. Values are presented as mean ± standard deviation. Experimental results were compared with published data.

	Alpha parameter	SF _{fit} (2 Gy)
Exp	2.100 ± 0.081	0.0150 ± 0.0031
Elgqvist et al., 2016 ¹⁸	2.01 ± 0.03	0.0180 ± 0.0036
Nilsson et al., 2015 ¹⁹	1.86 ± 0.94	0.02 ± 0.01
Stenerlöw et al., 1995 ²⁰	0.731 ± 0.187	0.232 ± 0.017

Radiobiological results

The cellular response to alpha radiation was shown in the form of the survival fraction of DU145 cells after exposure to 2 Gy. Value of the α parameter was estimated after the mono-exponential curve was fitted to the experimental data. Results were presented in **Table 3** and compared with data published before.¹⁸⁻²⁰ Parameters obtained from the fitted curves analyzed with 3 σ criterion can be considered consistent with the data from Nilsson et al. only. Remaining alpha parameters read from LQ model curves fitted to the DU145 cells survival data differ by more than three standard deviations of the mean due to very low uncertainties or significant differences of results.

Scanned Petri dishes with stained colonies and all experimental data on the surviving fraction of DU145 cells as a function of the mean absorbed dose for the ionizing radiation is shown in Attachment A.

Summary

The study of the ranges of alpha particles in water was performed using three tools based on Monte Carlo simulation (Geant4, GATE and SRIM). While SRIM was written to calculate ion deposition profiles in materials exposed to energetic beams of ions in simple geometries, Geant4 and GATE, which is based on the Geant4 toolkit, offer possibility of modelling advanced irradiation facilities. Regardless of the capabilities of the tool used, the results are consistent and the calculation accuracy is similar for all frameworks.

Monte Carlo methods are useful to model radioactive sources, which are unique tools to study cellular response to radiation in radiobiological research. Dosimetric characteristics of these sources play an important role in experiment design and performance and it is often based on manufacturer description. However, as it was shown in our studies dose distribution should be verified based on either dosimetric measurements or Monte Carlo simulations validated with experimental data. In the case when measurements are not in agreement with the theoretical predictions, the new model of the ion generation in the source has to be created and used for preparing the experimental setup allowing to uniformly irradiate cells.

To compare the homogeneity of dose distributions simulated with Geant4 and measured with radiochromic films the gamma

index was used. The criterion most commonly used for the quality assurance of conformal plans in 3D radiotherapy planning is gamma 3%/3mm. Distributions are considered to be consistent if at least 95% of the points reach a gamma parameter value below 1. The comparison of simulated and measured dose distribution was performed by using a more stringent passing criteria 1%/1mm due to the high spatial resolution of Gafchromic film and because single cells were modelled in Geant4.²¹ At 3%/3mm gamma analysis, both distributions met the acceptance criterion (despite the differences visible in the images).

The techniques used to improve the homogeneity of the dose delivered to cells by high LET radiation as presumed proved to be valid. The calculated CV values for radiochromic films irradiated in absorber-based and dynamic setups are lower than for a film placed directly on the source, indicating that both methods would benefit the quality of cell irradiation. It is worth noting that the method of increasing the distance between the film and the source is more accessible and easier to apply than the method using a laboratory shaker. This makes it more attractive from the point of view of radiobiological experiments.

One should take into account that α particles emitted from radioactive sources with energies equal to 5 MeV, after passing through the air layer and Mylar foil used to cover cells or to seed them, reach energies characteristic of the Bragg peak ($E < 3$ MeV). In hadron radiotherapy, which is increasingly used in cancer therapy, the projectile energies are selected so that in the area of the irradiated tumor they also reach energies characteristic for the Bragg peak. Research on the biological effects of α -particle irradiation from radioactive sources allows planning of hadron radiotherapy processes.

This work shows that irradiating cell colonies with a uniform dose of α radiation from a flat source is not easy, because the energy of particles emitted from the source is smeared due to deceleration in the medium (straggling), and additionally the energy losses in air and Mylar foil may differ significantly for different particle emission angles. Particles with different energies deliver different doses into biological material. The measured biological effects for a homogeneous dose with a value of D_0 and a heterogeneous dose with a value of $\langle D \rangle$, averaged over the range from D_{min} to D_{max} , may differ significantly even when $D_0 = \langle D \rangle$. Our preliminary analysis

indicates that the survival fraction SF determined for the averaged dose $SF(<D>)$ is overestimated compared to the survival fraction for the uniform dose $SF(D_0)$. Hence the attention paid in this work to examining dose distributions and their homogeneity.

Conclusions

The effective use of ionizing radiation in radiotherapy for cancer patients requires understanding the biological effects of radiation at the cellular level. For this purpose, radiobiological experiments are performed to investigate the average cellular response to different irradiation scenarios. However, the challenge is the use of densely ionizing radiation, which is often used in clinical practice because it spares healthy tissue during patient irradiation. This advantage is also a hindrance in the use of high LET radiation in radiobiological experiments, because the dose should be delivered uniformly to the cell monolayer. Computer tools come to the rescue, allowing us to simulate dose distribution and design an irradiation system to ensure that each

cell is treated the same, resulting in a reliable average response measured in the laboratory. Although access to various Monte Carlo codes is easy, it is worth remembering that the simulation results must be experimentally verified and only after confirmation of compliance can be used to plan radiobiological experiments, which is less expensive and time-consuming compared to laboratory work.

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Conflicts of Interest

The authors declare no conflict of interest.

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Attachment A: Survival data for irradiated DU145 cells

Mean plating efficiency was calculated based on the control DU145 cells (six replicates: three replicates from two independent experiments) and was equal to $(5.3 \pm 1.2)\%$.

Table A1. Mean SF data for DU145: experimental and fitted based on plating efficiency of control cells and surviving fraction of irradiated cells based on LQ model. The uncertainty of the SF_{fit} was calculated using the propagation of uncertainty. It was assumed that the uncertainty of dose is 0.1

Dose	SF (0 Gy)	SF (0.16 Gy)	SF (0.32 Gy)	SF (0.65 Gy)	SF (1.31 Gy)	SF (2 Gy)
SF_{exp}	1.0 ± 0.0	0.461 ± 0.046	0.315 ± 0.096	0.1835 ± 0.0085	0.052 ± 0.014	0.040 ± 0.014
SF_{fit}	1.0 ± 0.0	0.71 ± 0.15	0.51 ± 0.11	0.255 ± 0.054	0.064 ± 0.013	0.0150 ± 0.0031

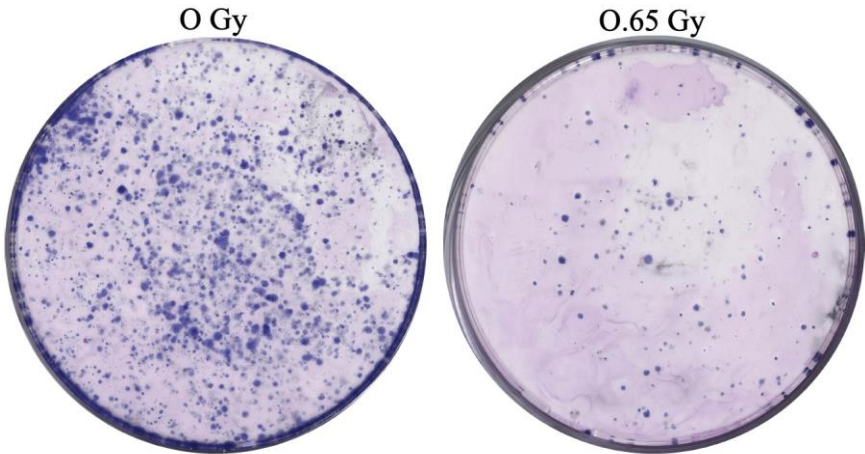


Figure A1. Representative images of stained colonies for prostate cancer cells DU145 and variable doses for alpha radiation.